

LONGSTAFF BLUFF, NU, Canada

WMO: 710910

Lat: 68.899N

Lon: 75.141W

Elev: 528

StdP: 14.42

Time Zone: -5.00 (NAE)

Period: 82-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-41.4	-38.5	-41.6	0.5	-33.5	-40.5	0.5	-33.4	42.9	-1.4	37.8	-4.6	7.5	90	1.105

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	59.6	50.1	56.7	48.1	53.7	46.4	50.9	58.9	48.6	55.5	46.5	52.3	9.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
45.5	45.8	53.5	43.7	42.7	51.1	41.9	39.9	48.7	20.9	59.3	19.7	55.9	18.6	52.3	55.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.4	31.0	26.8	-44.9	64.2	4.6	3.7	-48.3	66.8	-51.0	69.0	-53.5	71.1	-56.9	73.9		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-19.2	-20.9	-15.5	-0.4	18.6	35.7	46.7	42.7	32.3	18.2	2.1	-10.7	(6)
(7)		DBStd	25.63	12.26	11.61	12.01	10.79	8.79	6.20	6.05	5.59	5.56	8.73	11.38	13.05	(7)
(8)		HDD50	14287	2144	1985	2031	1512	972	430	138	239	532	987	1436	1881	(8)
(9)		HDD65	19713	2609	2405	2496	1962	1437	879	566	693	982	1452	1886	2346	(9)
(10)		CDD50	49	0	0	0	0	0	1	37	11	0	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	12.4	11.7	11.5	11.8	12.4	12.5	11.6	10.9	11.8	13.1	14.5	14.5	12.3	(14)
(15)	Precipitation	PrecAvg	12.0	0.5	0.5	0.6	0.7	0.7	1.0	1.4	2.0	1.5	1.5	0.9	0.7	(15)
(16)		PrecMax	19.4	1.5	1.9	2.3	2.7	1.6	2.7	3.7	3.3	2.9	3.3	3.5	1.6	(16)
(17)		PrecMin	5.2	0.0	0.0	0.0	0.1	0.0	0.3	0.4	0.2	0.3	0.3	0.1	0.2	(17)
(18)		PrecStd	3.3	0.4	0.4	0.5	0.6	0.4	0.5	0.7	0.7	0.7	0.7	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.7	16.5	21.3	29.3	39.8	55.7	64.8	61.5	47.7	33.5	28.6	26.4	(19)
(20)			MCWB	21.5	15.4	20.7	28.6	36.7	47.4	52.3	49.8	40.9	32.0	27.8	24.7	(20)
(21)		2%	DB	10.8	7.9	13.5	22.7	35.0	51.2	61.3	57.5	44.2	31.8	25.6	19.5	(21)
(22)			MCWB	10.1	7.3	12.8	21.3	33.4	44.4	50.8	49.2	39.4	31.6	24.8	18.6	(22)
(23)		5%	DB	5.0	1.4	6.9	19.1	32.6	48.0	58.8	54.0	41.6	30.6	22.8	13.0	(23)
(24)			MCWB	4.2	0.7	6.2	17.9	31.9	42.3	49.7	47.6	38.6	30.2	22.3	12.2	(24)
(25)		10%	DB	-1.3	-4.7	1.8	15.0	30.9	45.0	56.2	51.1	39.7	28.8	18.7	7.4	(25)
(26)			MCWB	-1.9	-5.3	1.3	14.1	30.3	40.4	47.9	44.5	37.4	28.1	18.0	6.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	12.5	19.0	22.4	37.6	49.6	53.8	48.4	42.4	32.1	26.0	22.6	(27)
(28)			MCD B	20.6	13.1	19.6	23.1	39.7	56.6	63.1	53.8	45.9	32.1	26.2	23.0	(28)
(29)		2%	WB	7.2	5.1	7.9	19.4	34.1	45.4	52.1	45.6	40.7	30.5	23.0	14.2	(29)
(30)			MCD B	7.8	5.6	8.5	20.6	36.0	51.8	60.8	49.7	42.9	30.6	23.6	14.8	(30)
(31)		5%	WB	1.4	-0.4	1.9	16.7	32.1	43.4	50.7	43.8	38.8	28.7	18.2	7.7	(31)
(32)			MCD B	1.9	0.2	2.4	17.5	33.0	49.1	58.8	47.2	41.1	29.0	18.8	8.4	(32)
(33)		10%	WB	-3.5	-6.2	-3.5	12.9	29.4	41.1	48.9	42.4	37.1	26.8	14.2	3.3	(33)
(34)			MCD B	-2.8	-5.8	-3.0	13.7	29.8	45.7	56.1	45.6	39.1	27.4	15.0	3.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.1	9.9	10.2	10.2	7.8	7.0	9.5	7.9	5.5	7.0	9.7	9.9	(35)
(36)			MCD BR	13.8	14.7	13.2	11.6	7.7	11.1	12.1	11.4	8.2	6.2	10.9	12.1	(36)
(37)			MCWBR	13.5	16.0	12.5	11.9	6.4	7.1	6.8	6.1	5.3	6.6	14.7	11.2	(37)
(38)		5% WB	MCD BR	13.2	15.6	12.0	11.7	6.9	10.8	11.5	8.9	7.4	6.1	13.3	11.0	(38)
(39)			MCWBR	12.8	15.2	12.0	11.5	6.3	7.3	7.2	5.9	5.5	6.4	13.3	10.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.158	0.197	0.244	0.299	0.320	0.326	0.341	0.319	0.286	0.231	0.154	N/A	(40)	
(41)		taud	1.860	2.072	2.199	2.078	2.126	2.325	2.429	2.528	2.511	2.129	1.908	N/A	(41)	
(42)		Ebn at Noon	22	194	254	265	272	276	267	263	243	168	20	0	(42)	
(43)		Edh at Noon	2	13	23	36	39	33	29	24	18	12	1	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6	131	654	1375	1901	1971	1644	1112	580	183	20	0	(44)	
(45)		RadStd	1	18	32	63	124	230	145	95	52	25	3	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +1.06	(e) +2.27	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) -432	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (2 neighbors)	+1.06	+2.27	N/A	N/A	N/A	N/A	-432	N/A	N/A	N/A	(47)

Nomenclature: See separate page